

Your recent learnings

When you left 2 days ago, you worked on Transforming DataFrames, chapter 1 of the course Data Manipulation with pandas. Here is what you covered in your last lesson:

You learned about adding new columns to a DataFrame in pandas, an essential skill for data manipulation and feature engineering. This process allows you to enrich your data based on existing columns or calculations. Here are the key points:

- **Creating New Columns:** You discovered how to create new columns in a DataFrame by performing operations on existing columns. For example, to convert a dog's height from centimeters to meters, you would use `DataFrame["new_column"] = DataFrame["existing_column"] / 100 .`
- **Calculating BMI for Dogs:** You applied this concept to calculate the Body Mass Index (BMI) for dogs, using the formula `BMI = weight_kg / (height_m)^2 .` This involved creating a new column where each entry is derived from two existing columns.
- **Combining Skills:** You combined various pandas skills to identify tall, skinny dogs. This involved subsetting rows based on BMI, sorting by height, and selecting specific columns to view the result.
- **Real-world Application:** You practiced these skills with a real dataset by adding a column to track the total number of homeless individuals and families, and another to calculate the proportion of homeless individuals per state. The code snippet below illustrates how to add these columns:

```
# Add total col as sum of individuals and family_members
homelessness["total"] = homelessness["individuals"] + homelessness["family_members"]

# Add p_individuals col as proportion of total that are individuals
homelessness["p_individuals"] = homelessness["individuals"] / homelessness["total"]
```

- **Advanced Manipulation:** Finally, you learned to combine data manipulation techniques to answer complex questions, such as identifying states with a high number of homeless individuals per capita.